

measure 0

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1 Notation

Let $f : A \rightarrow [0, \infty]$. Define $\sum_{x \in A} f(x) = \sup \{ \sum_{x \in B} f(x) : B \subseteq A \text{ is finite} \}$. Note that $\emptyset \subseteq A$, so $\sum_{x \in A} f(x)$ is well-defined and $\sum_{x \in A} f(x) \geq 0$.

Sps A, B , $f : A \cup B \rightarrow [0, \infty]$. Then $\sum_{x \in A \cup B} f(x) \leq \sum_{x \in A} f(x) + \sum_{x \in B} f(x)$.

Proof: Let $S \subseteq A \cup B$ be finite. Let $S_A = A \cap S$ and $S_B = B \cap S$. Then $S = S_A \cup S_B$. We have

$$\sum_{x \in S_A \cup S_B} f(x) + \sum_{x \in S_A \cap S_B} f(x) = \sum_{x \in S_A} f(x) + \sum_{x \in S_B} f(x),$$

so $\sum_{x \in S_A} f(x) + \sum_{x \in S_B} f(x) \geq \sum_{x \in S_A \cup S_B} f(x)$. S_A, S_B are finite, so $\sum_{x \in A} f(x) + \sum_{x \in B} f(x) \geq \sum_{x \in S_A} f(x) + \sum_{x \in S_B} f(x) \geq \sum_{x \in S} f(x)$ for all finite $S \subseteq A \cup B$, so $\sum_{x \in A \cup B} f(x) \leq \sum_{x \in A} f(x) + \sum_{x \in B} f(x)$.

Let $a : N \rightarrow [0, \infty]$. I claim that $\sum_{n \in N} a_n = \sum_{n=1}^{\infty} f(n)$.

Proof: The partial sums $\sum_{n=1}^N a_n$ are increasing, so $\sum_{n=1}^{\infty} f(n) = \sup \{ \sum_{n=1}^N a_n : N \in N \}$.

We have $\sum_{n=1}^N a_n = \sum_{n \in [N]} a_n$, so $\sum_{n=1}^{\infty} f(n) \leq \sum_{n \in N} f(n)$. Conversely, if $S \subseteq N$ is finite, there exists $N \in N$ such that $S \subseteq [N]$, so $\sum_{n \in S} f(n) \leq \sum_{n \in [N]} f(n) = \sum_{n=1}^N f(n) \leq \sum_{n=1}^{\infty} f(n)$, so $\sum_{n \in N} f(n) \leq \sum_{n=1}^{\infty} f(n)$.

Let $a : N^2 \rightarrow [0, \infty]$. Then

$$\sum_{(n,m) \in N^2} a(n,m) = \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} a(n,m) = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} a(n,m).$$

Let A, B be finite. Then

$$\begin{aligned} \sum_{x \in A} f(x) + \sum_{x \in B} f(x) &= \sum_{x \in A} f(x) + \sum_{x \in B-A} f(x) + \sum_{x \in A \cap B} f(x) \\ &= \sum_{x \in A \cup B} f(x) + \sum_{x \in A \cap B} f(x). \end{aligned}$$

Let A, B be disjoint finite. Then $\sum_{x \in A} f(x) + \sum_{x \in B} f(x) = \sum_{x \in A \cup B} f(x)$. Let $h_1 : [n] \rightarrow A$ and $h_2 : [m] \rightarrow B$ be bijections. Define $h_3 : [n+m] \rightarrow A \cup B$ as usual by

$$h_3(i) = \begin{cases} h_1(i) & \text{if } 1 \leq i \leq n, \\ h_2(i-n) & \text{if } n+1 \leq i \leq n+m. \end{cases}$$

I claim that h_3 is a bijection. Sps $h_3(i) = h_3(j)$. If $1 \leq i \leq n$ and $n+1 \leq j \leq n+m$, then $h_3(i) \in A$ and $h_3(j) \in B$, absurd. Similarly, we cannot have $n+1 \leq i \leq n+m$ and $1 \leq j \leq n$. Assume $1 \leq i, j \leq n$. Then $h_1(i) = h_1(j) \rightarrow i = j$. Similarly, if $n+1 \leq i, j \leq n+m$, then $i = j$.

$h_3([n+m]) = h_1([n]) \cup h_2([m]) = A \cup B$, so h_3 is surjective. Hence, h_3 is a bijection.

Then

$$\begin{aligned} \sum_{x \in A \cup B} f(x) &= \sum_{i=1}^{n+m} h_3(i) = \sum_{i=1}^n h_3(i) + \sum_{i=n+1}^{n+m} h_3(i) \\ &= \sum_{i=1}^n h_1(i) + \sum_{i=1}^m h_2(i) = \sum_{x \in A} f(x) + \sum_{x \in B} f(x). \end{aligned}$$

Let A, B be disjoint (need not be finite). Then $\sum_{x \in A} f(x) + \sum_{x \in B} f(x) = \sum_{x \in A \cup B} f(x)$.

Proof: We have $\sum_{x \in A} f(x) + \sum_{x \in B} f(x) \geq \sum_{x \in A \cup B} f(x)$. Suppose $S_A \subseteq A, S_B \subseteq B$ are finite. Since A, B are disjoint, so are S_A, S_B . Then

$$\sum_{x \in S_A} f(x) + \sum_{x \in S_B} f(x) = \sum_{x \in S_A \cup S_B} f(x) \leq \sum_{x \in A \cup B} f(x).$$

Fix S_B and take the supremum over S_A . Then take the supremum over S_B . Then $\sum_{x \in A} f(x) + \sum_{x \in B} f(x) \leq \sum_{x \in A \cup B} f(x)$.

Claim: Let $\{A_n\}_{n \in \mathbb{N}}$ be a collection of sets such that $A_n \subseteq A_{n+1}$ for all $n \in \mathbb{N}$ and let $A = \bigcup_{n \in \mathbb{N}} A_n$. Sps $f : A \rightarrow [0, \infty]$. I claim that $\sum_{x \in A} f(x) = \sup_{n \in \mathbb{N}} \sum_{x \in A_n} f(x)$.

Proof: $\sum_{x \in A} f(x) \geq \sum_{x \in A_n} f(x)$ for all $n \in \mathbb{N}$, so $\sum_{x \in A} f(x) \geq \sup_{n \in \mathbb{N}} \sum_{x \in A_n} f(x)$.

Suppose $S \subseteq A$ is finite. Enumerate $h : [n] \rightarrow S$. For each $i \in [n]$, $h(i) \in S$ and $h(i) \in A_{b(i)}$ for some $b(i) \in N$. Let $M = \max_{i \in [n]} b(i)$, so $h(i) \in A_M$ for all $i \in [n]$. Then $\sum_{x \in S} f(x) \leq \sum_{x \in A_M} f(x) \leq \sup_{n \in N} \sum_{x \in A_n} f(x)$. Hence, $\sum_{x \in A} f(x) \leq \sup_{n \in N} \sum_{x \in A_n} f(x)$.

Note that $N^2 = \bigcup_{n \in N} [n] \times N$, so

$$\begin{aligned} \sum_{(n,m) \in N^2} a(n,m) &= \sup_{N \in N} \sum_{(n,m) \in [N] \times N} a(n,m) \\ &= \sup_{N \in N} \sum_{n=1}^N \sum_{m \in N} a(n,m) = \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} a(n,m). \end{aligned}$$

Let A, B be sets and $f : A \times B \rightarrow [0, \infty]$. Then

$$\sum_{(x,y) \in A \times B} f(x,y) = \sum_{x \in A} \sum_{y \in B} f(x,y) = \sum_{y \in B} \sum_{x \in A} f(x,y).$$

Proof: Let $C = \{(x,y) \in A \times B : f(x,y) > 0\}$. Sps C is uncountable. Note that $C \subseteq \pi_1(C) \times \pi_2(C)$, so at least one of $\pi_1(C), \pi_2(C)$ is uncountable. Sps $\pi_1(C)$ is uncountable. Then

$$\sum_{x \in A} \sum_{y \in B} f(x,y) \geq \sum_{x \in \pi_1(C)} \sum_{y \in B} f(x,y) = \infty,$$

since $\sum_{y \in B} f(x,y) > 0$ for all $x \in \pi_1(C)$.

Sps $\pi_1(C)$ is countable and $\pi_2(C)$ is uncountable. Then

$$\begin{aligned} \sum_{x \in A} \sum_{y \in B} f(x,y) &\geq \sum_{x \in \pi_1(C)} \sum_{y \in \pi_2(C)} f(x,y) = \sum_{(x,y) \in \pi_1(C) \times \pi_2(C)} f(x,y) \\ &= \sum_{(x,y) \in A \times B} f(x,y) = \infty. \end{aligned}$$

Else, assume C is countable, so $\pi_1(C), \pi_2(C)$ are countable. Then

$$\begin{aligned} \sum_{(x,y) \in A \times B} f(x,y) &= \sum_{(x,y) \in \pi_1(C) \times \pi_2(C)} f(x,y) = \sum_{x \in \pi_1(C)} \sum_{y \in \pi_2(C)} f(x,y) \\ &= \sum_{x \in \pi_1(C)} \sum_{y \in B} f(x,y) = \sum_{x \in A} \sum_{y \in B} f(x,y). \end{aligned}$$

2 Measure Theory

Let $A_1, A_2 \subseteq \mathbb{R}^d$ be elementary. Then $A_1 \cup A_2$ is elementary. Let $A_1 = \bigcup_{i \in I_1} B_i$ and $A_2 = \bigcup_{i \in I_2} B_i$. Then $A_1 \cup A_2 = \bigcup_{i \in I_1 \cup I_2} B_i$ is elementary.

$$A_1 \cap A_2 = \bigcup_{i \in I_1} B_i \cap \bigcup_{j \in I_2} B_j = \bigcup_{(i,j) \in I_1 \times I_2} B_i \cap B_j.$$

Sps $B = \prod_{i=1}^d I_i$ and $B' = \prod_{i=1}^d I'_i$. Then $B \cap B' = \prod_{i=1}^d I_i \cap I'_i$. $I_i \cap I'_i$ is an interval, so $B \cap B'$ is a box. Hence, $A_1 \cap A_2$ is elementary.

$$\begin{aligned} A_1 - A_2 &= \bigcup_{i \in I_1} B_i - \bigcup_{j \in I_2} B_j = \bigcup_{i \in I_1} \left(B_i - \bigcup_{j \in I_2} B_j \right) \\ &= \bigcup_{i \in I_1} \bigcap_{j \in I_2} (B_i - B_j) \end{aligned}$$

Note that

$$A \times B \times -A' \times B' = (A \times (B - B')) \cup ((A - A') \times B),$$

and in general

$$\prod_{i=1}^n A_i - \prod_{i=1}^n A'_i = \bigcup_{i \in [n]} \left(\prod_{j=1}^{i-1} A_j \times (A_i - A'_i) \times \prod_{j=i+1}^n A_j \right),$$

so the difference of two boxes is elementary. Hence, $\bigcap_{j \in I_2} (B - B_j)$ is elementary (since I_2 is finite), and therefore $\bigcup_{i \in I_1} \bigcap_{j \in I_2} (B - B_j)$ as well.

Corollary: $E \Delta F = (E - F) \cup (F - E)$ is elementary.

Let $A \subset \mathbb{R}^d$ and define $x + A = \{x + a : a \in A\}$. Let B be a box. Then $x + B$ is a box.

Proof: Write $x = (x_i)_{i \in [d]}$. Sps $B = \prod_{i=1}^d I_i$. Sps $y \in x + B$. Then $y_i = x_i + b_i$, where $b_i \in I_i$, so $y_i \in x_i + I_i$, which is an interval, so $y \in \prod_{i=1}^d (x_i + I_i)$. Conversely, if $y \in \prod_{i=1}^d (x_i + I_i)$, then $y = x + b$ for some $b \in B$. Hence, $x + \prod_{i=1}^d I_i = \prod_{i=1}^d (x_i + I_i)$.

It follows that if E is elementary, so is the translate $x + E$.

Let I be an interval and $\mathcal{P}$ be a partition of I into intervals. Then $|I| = \sum_{J \in \mathcal{P}} |J|$.

Proof: Define $f_J : \mathbb{R} \rightarrow \mathbb{R}$ by $f_J(x) = 1$ if $x \in J$ and $f_J(x) = 0$ otherwise. If $x \notin I$, then $x \notin J$, so $f_J(x) = 0$. Write $J \subseteq [a, b]$. Hence,

$$\int_{-\infty}^{\infty} \sum_{J \in \mathcal{P}} f_J(x) dx = \int_a^b \sum_{J \in \mathcal{P}} f_J(x) dx = \sum_{J \in \mathcal{P}} \int_a^b f_J(x) dx = \sum_{J \in \mathcal{P}} |J|.$$

On the other hand, $\sum_{J \in \mathcal{P}} f_J(x) = 1$ for all $x \in J$, so

$$\int_a^b \sum_{J \in \mathcal{P}} f_J(x) dx = \int_a^b dx = |J|,$$

as desired.

Let $\mathcal{P}$ be a partition of the d -dimensional box B . Write $B = I \times B'$, where B' is a box of dimension $d - 1$. For each $C \in \mathcal{P}$, write $C = K \times C'$. Define $f_C : \mathbb{R} \rightarrow \mathbb{R}$ by $f_C(x) = |C'|$ if $x \in K$ and $f_C(x) = 0$ otherwise. Consider

$$\int_I \sum_{C \in \mathcal{P}} f_C(x) = \sum_{C \in \mathcal{P}} \int_I f_C(x) = \sum_{C \in \mathcal{P}} |K| \cdot |C'| = \sum_{C \in \mathcal{P}} |C|$$

and on the other hand, for $x \in I$, B' is partitioned into boxes by $\pi_{>1}((x \times B') \cap \mathcal{P})$. By the inductive hypothesis,

$$\sum_{C \in \mathcal{P}} f_C(x) = \sum_{C' \in \pi_{>1}((x \times B') \cap \mathcal{P})} |C'| = |B'|,$$

so

$$\int_I \sum_{C \in \mathcal{P}} f_C(x) = \int_I |B'| = |I| \cdot |B'| = |B|.$$

Hence, if $\mathcal{P} \prec \mathcal{Q}$ are partitions of E into boxes, then $\sum_{B \in \mathcal{P}} |B| = \sum_{B \in \mathcal{Q}} |B|$.

Sps $\mathcal{P}, \mathcal{Q}$ are partitions of E into boxes. Let $\mathcal{P} \# \mathcal{Q} = \{B \cap C : B \in \mathcal{P}, C \in \mathcal{Q}\}$. Since B, C are boxes, so is $B \cap C$. $\mathcal{P} \# \mathcal{Q} \prec \mathcal{P}, \mathcal{Q}$.

I claim that every elementary set E can be partitioned into boxes.

Consider $n = 1$. Sps $J \in \mathcal{P}$, so $J = [a, b], [a, b), (a, b], (a, b)$. Ignore J if $a = b$ and J is not an interval of the first form, so $J \neq \emptyset$. Let X denote the set of a 's and b 's.

Since $\mathcal{P}$ is finite, so is X . Assume $I \neq \emptyset$, so $\mathcal{P} \neq \emptyset$ and $X \neq \emptyset$. Let $h : [n] \rightarrow X$ be the increasing enumeration of X .

Let $Y = \{x \in X : x \in I\}$. Let $U = \bigcup_{i \in [n-1]} (h(i), h(i+1))$. I claim that $Y \cup U = I$. Sps $I = [a, b], [a, b)$. There exists $J \in \mathcal{P}$ such that $a \in J$. Then $J \neq (c, d), (c, d]$, so $J = [a, c], [a, c]$, so $a \in Y$.

Let X have linear order and $S \subseteq X$ be finite. Then there exists a bijection $f : [n] \rightarrow S$ such that f is increasing.

Proof: If $S = \emptyset$, take f to be the empty function. Sps the result for n and

let $|S| = n + 1$.

Sps the result for n .